OR Central Exercise

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Integer Programming

29 May 2026

Question 1

	Profit per hectare	
	With good soil (g)	With bad soil (b)
Wheat (W)	10	5
Corn (C)	12	4
Potatoes (P)	15	6
Rye (R)	7	7

→ table for c_{pq} values

$x_{p,q,t} \geq 0$, for all $p \in P := \{W, C, P, R\}$, $q \in Q := \{g, b\}$, and $t \in T := \{1, 2, \dots, 15\}$.

part a)

Specify the objective function to be maximized as a function of $x_{p,q,t}$.

- define $c_{p,q}$: ~~profit from~~ profit from p harvested from quality q soil.
- write down the open expression

$$\sum_{t \in T} 10 x_{Wgt} + 5 x_{Wbt} + \dots$$

$$\sum_{t \in T} \sum_{p \in P} \sum_{q \in Q} c_{pq} x_{pqt}$$

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part b) (i)

No more than 1000 hectares of land may be used for crop cultivation every year.

$$\sum_{p \in P} \sum_{q \in Q} x_{pqt} \leq 1000 \qquad \forall t \in T$$

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part b) (ii)

If an agricultural product was cultivated in each of the two preceding years on at most 50 hectares, then it may be cultivated on good soil in year t; otherwise, it can only be grown on poor soil. Assume that no crops were cultivated before year $t = 1$.

Define $A_{p,t} \in \{0, 1\}$: $A_{p,t} = 1$ if p is grown on more than 50 hectares in t.

$$x_{pgt-1} + x_{pbt-1} - 50 \leq A_{pt} \cdot 1000 \quad \forall t = 2..15$$

$$x_{pgt} \leq 1000[1 - A_{pt-1}] \;\; \forall t \geq 2 \quad \& \quad x_{pgt} \leq 1000[1 - A_{pt-2}] \;\; \forall t \geq 3$$

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part b) (iii)

The total area of potatoes cultivated over all years may not exceed 1500 hectares.

$$\sum_{t \in T} \sum_{q \in Q} x_{pqt} \leq 1500$$

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part b) (iv)

If rye has been cultivated in 3 consecutive years, no rye may be cultivated in the following year.

Define $R_t \in \{0, 1\}$: $R_t = 1$ if rye has been grown in t.

$$x_{Rgt} + x_{Rbt} \leq 1000 R_t \quad \forall t \in T$$

$$[-R_{t-1} - R_{t-2} - R_{t-3} + 3] \geq R_t$$

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part b) (v)

If wheat and rye are cultivated in a year, at least one other agricultural product must be cultivated on an area of at least 300 hectares.

$1 - P_t$

Define $R_t, W_t \in \{0, 1\}$ as before: $W_t = 1$ if wheat has been grown in t.

Define $P_t \in \{0, 1\}$: $P_t = 1$ if potatoes should be grown in t on at least 300.

$\{0,1,2\}$

$X_{wgt} + X_{wbt} \leq 1000 W_t \quad \forall t \in T \qquad R_t \,\&\, W_t = 1$

$X_{pgt} + X_{pbt} \geq 300 P_t \quad \forall t \in T \qquad X_{cgt} + X_{cbt} \geq 300[1 - P_t + R_t + W_t - 2]$

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part b) (v) - alternative solution

If wheat and rye are cultivated in a year, at least one other agricultural product must be cultivated on an area of at least 300 hectares.

Define $P_t, C_t \in \{0, 1\}$: $C_t = 1$ if corn should be grown in t on at least 300.

$$1000W_t \geq x_{W,g,t} + x_{W,b,t} \qquad 1000R_t \geq x_{R,g,t} + x_{R,b,t} \qquad \forall t \in T,$$

$$W_t + R_t - 1 \leq P_t + C_t \quad \forall t \in T$$

$$300\, P_t \leq X_{pgt} + X_{pbt} \ \forall t, \quad 300\, C_t \leq X_{cgt} + X_{cbt} \ \forall t$$

Question 2

$I = \{1, \ldots, 15\}$ possible school sites, $c_i \in \mathbb{R}_{>0}$ capacity, and $f_i \in \mathbb{R}_{>0}$ cost.
$J = \{1, \ldots, 800\}$ students, $s_{i,j} \in \mathbb{R}_{>0}$ preference scores for all $i \in I$, $j \in J$.
$y_i \in \{0, 1\}$: 1 if $i \in I$ is built and 0 otherwise.
$x_{i,j} \in \{0, 1\}$: 1 if $j \in J$ is assigned to $i \in I$ and 0 otherwise.

part a)

Each student must be assigned to exactly one school.

$$\sum_{i \in I} x_{ij} = 1 \qquad \forall j \in J$$

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part b)

The total number of students assigned to each school cannot exceed the school's capacity.

$$\sum_{j \in J} x_{ij} \leq c_i \quad \forall i \in I$$

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part c)

A student may be assigned to a school only if the school is built.

- for each student $\rightarrow$ $x_{ij} \leq y_i \quad \forall i \in I \; \forall j \in J$
- for all students

$$\sum_{i \in I} x_{ij} \leq y_i \cdot M \quad \forall i \in I$$

M: big number

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part d)

$d_{i,i'} \in \mathbb{R}_{>0}$ denotes the distance between $i, i' \in I$ with $i \neq i'$. The distance between any two built schools must be at least 5 km.

Define $z_{i,i'} \in \{0, 1\}$: $z_{i,i'} = 1$ if both i and i' are open.

$$y_i + y_{i'} - 1 \leq z_{ii'} \quad \forall i, i' \in I : i \neq i'$$

$$d_{ii'} \geq 5 z_{ii'} \quad \forall i, i' \in I : i \neq i'$$

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part e)

$A_i \in \{0, 1\}$ indicates $i \in I$ is located in zone A ($A_i = 1$) or in zone B ($A_i = 0$). If more than 200 students are assigned to A, then at least 300 students must be assigned to B.

Define $y \in \{0, 1\}$: $y = 1$ if more than 200 students are assigned to A.

$$\sum_{i \in I} \sum_{j \in J} x_{ij} A_i - 200 \leq 600 y$$

$$\sum_{i \in I} \sum_{j \in J} x_{ij} (1 - A_i) \geq 300 y$$

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part f)

Formulate a single objective function that minimizes total construction costs while maximizing total preference scores of students.

$$\min \sum_{i \in I} f_i y_i \quad \& \quad \max \sum_{j \in J} \sum_{i \in I} s_{ij} x_{ij}$$

$$\max \; w_p \left[\sum_{j \in J} \sum_{i \in I} s_{ij} x_{ij} \right] - \left[\sum_{i \in I} f_i y_i \right] w_c \qquad w_p = w_c > 0$$